

4.4 How far? – entropy

Students will be assessed on their ability to:

- a demonstrate an understanding that, since endothermic reactions can occur spontaneously at room temperature, enthalpy changes alone do not control whether reactions occur
- b demonstrate an understanding of entropy in terms of the random dispersal of molecules and of energy quanta between molecules

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- c demonstrate an understanding that the entropy of a substance increases with temperature, that entropy increases as solid → liquid → gas and that perfect crystals at zero kelvin have zero entropy
 - d demonstrate an understanding that the standard entropy of a substance depends mainly on its physical state but also on its complexity
 - e demonstrate an understanding that reactions occur due to chance collisions, and that one possible ordered arrangement, eg in a crystalline solid, can be rearranged into many possible disordered arrangements, eg in a solution, so the probability of disorder is greater than order
 - f interpret the natural direction of change as being in the direction of increasing total entropy (positive entropy change), eg gases spread spontaneously through a room
 - g carry out experiments and relate the results to disorder and enthalpy changes including:
 - i dissolving a solid, eg adding ammonium nitrate crystals to water
 - ii gas evolution, eg reacting ethanoic acid with ammonium carbonate
 - iii exothermic reaction producing a solid, eg burning magnesium ribbon in air
 - iv endothermic reaction of two solids, eg mixing solid barium hydroxide, $\text{Ba(OH)}_2 \cdot 8\text{H}_2\text{O}$ with solid ammonium chloride
 - h demonstrate an understanding that the entropy change in any reaction is made up of the entropy change in the system added to the entropy change in the surroundings, summarised by the expression:

$$\Delta S_{\text{total}} = \Delta S_{\text{system}} + \Delta S_{\text{surroundings}}$$
 - i calculate the entropy change in the system for a reaction, ΔS_{system} , given entropy data
 - j use the expression $\Delta S_{\text{surroundings}} = \frac{-\Delta H}{T}$ to calculate the entropy change in the surroundings and hence ΔS_{total}